Year 13 Maths - Sequences and series

Similar questions - Arithmetic progressions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The 12th term of an arithmetic progression is 38 and the sum of its first 8 terms is 124.

(a) Find the first term and the common difference. **[4]**

(b) Find the least value of n for which the sum of the first n terms exceeds 1000. **[4]**

(c) Find $\sum_{r=1}^{40} u_r$. **[2]**

2 The sum of the first n terms of an arithmetic progression is $S_n = 2n^2 + 3n$.

(a) Find the first term and the second term, and hence the common difference. **[3]**

(b) Show that the nth term is $4n + 1$. **[3]**

(c) Find the least n for which the nth term exceeds 200. **[2]**

(d) Find the sum of the terms from the 20th to the 40th inclusive. **[3]**

Answers

Question 1

(a) $a+11d=38$ and $\frac{8}{2}(2a+7d)=124$, so $2a+7d=31$. From the first, $2a+22d=76$; subtracting: $15d=45$, so $d=3$ and $a=5$

(b) $S_n=\frac{n}{2}(10+3(n-1))=\frac{n}{2}(3n+7)>1000$, so $3n^2+7n-2000>0$. $S_{25}=\frac{25}{2}(82)=1025>1000$ and $S_{24}=\frac{24}{2}(79)=948<1000$. So $n=25$

(c) $S_{40}=\frac{40}{2}(10+39\times 3)=20\times 127=2540$

Question 2

(a) $u_1=S_1=5$. $u_2=S_2-S_1=14-5=9$. $d=4$

(b) $u_n=S_n-S_{n-1}=2n^2+3n-2(n-1)^2-3(n-1)=4n+1$ (and $u_1=5$ fits)

(c) $4n+1>200$ gives $n>49.75$, so $n=50$

(d) $S_{40}-S_{19}=(3200+120)-(722+57)=3320-779=2541$